

Patient Shortage in Histopathology: Within-Patient Directions Fail by Orientation, Not by Weight

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Abstract

Adding the within-patient complement of a five-patient cohort to cohort whitening recovers only 17 percent of the 2.66-point gap between cohort-basis and pool-basis whitening on TCGA-UT, although the added directions span almost all of the pool’s missing between-patient variance. The preceding study attributed this shortfall to the weights: the added directions carry patch-level variance, not patient-level variance. This report tests that explanation with oracle information, as a mechanism study rather than a remedy. Each within-patient direction receives the between-patient variance of the full training pool along it; two brackets place the same oracle weights on random complement directions and replace the within-patient directions by the eigenbasis of the pool’s own between-patient covariance outside the cohort basis. The explanation fails. At five patients per class, oracle weights on the within-patient directions reach the same accuracy as the tuned patch-level weights, -0.01 points on TCGA-UT (95% interval: -0.11 to 0.10) and -0.13 on BRACS (-0.73 to 0.42), because the patch-level ranking of these directions already matches their patient-level ranking (Spearman 0.999 and 0.98) and the oracle trace on TCGA-UT, 0.30 times the cohort trace, equals the best fixed added trace of the source study. The pool’s own complement eigenbasis, which spans the same subspace on TCGA-UT, gains 1.67 points more (1.37 to 1.99) and closes 80 percent of the gap; on BRACS it gains 2.01 points more (0.74 to 3.38) and closes 77 percent. The within-patient directions therefore reach the right subspace but are oriented wrongly within it: attaching the correct variance to each of them cannot recover the pool’s concentration of between-patient variance on few directions, and the cohort has no patient-level information that could.

1 Motivation

Cohort whitening contracts the features along the eigenbasis of the between-patient covariance of the labelled cohort and gains 2.26 percentage points over the ordinary classifier at five patients per class on TCGA-UT (Queisler, 2026c). The same whitening along the between-patient basis of the full training pool gains 4.92 points, but it reads every training patient and serves as a mechanism reference. Queisler (2026d) traced almost all of the 2.66 -point difference to directions outside the cohort basis: correcting the cohort’s eigenvalues within its basis is worth only 0.08 points.

Queisler (2026e) tried to reach those directions from the cohort’s own patches. It added the within-patient covariance, projected off the cohort basis, to the cohort’s between-patient covariance at a tuned total size. The added directions located the missing span well: at five TCGA-UT patients the leading eight times r_B eigenvectors of the expanded covariance held 98 percent of the pool’s between-patient variance. Accuracy followed only weakly, with a gain of 0.45 points, a sixth of the gap. That report offered an explanation it could not test: the added directions carry patch-level variances, whereas whitening needs the patient-level variance along each direction, which the cohort’s few patient means do not estimate outside their own basis.

This report tests that explanation directly by giving the within-patient directions the weights they lack. The weights come from the training pool, so the study measures a mechanism and is

not a remedy. The question is: *if every within-patient direction carries the pool’s between-patient variance along it, does cohort whitening with the added directions close the gap to pool-basis whitening?* A positive answer would locate the bottleneck in the weights and motivate cohort-only estimates of patient-level strength. A negative answer, together with a bracket that also corrects the directions, would locate it in the directions themselves.

2 Estimator model

Write the patch features of class c as $\mathbf{x}_{ik} = \boldsymbol{\mu}_c + \mathbf{u}_i + \mathbf{e}_{ik}$, with class centre $\boldsymbol{\mu}_c$, patient offset $\mathbf{u}_i$, and within-patient residual $\mathbf{e}_{ik}$ (Queisler, 2026b). A cohort of G patients per class estimates the between-patient covariance from the deviations of its patient means from their class means, pooled over all C classes, $\hat{\boldsymbol{\Sigma}}_B = \mathbf{U}^\top \text{diag}(\boldsymbol{\lambda}) \mathbf{U}$, where the $r_B \leq C(G - 1)$ rows of $\mathbf{U}$ are its eigenvectors with non-negligible eigenvalues $\boldsymbol{\lambda}$. Every arm of this study whitens by

$$\mathbf{x} \mapsto \mathbf{x}(\mathbf{I} + \kappa \mathbf{B}^\top \text{diag}(\boldsymbol{\nu}) \mathbf{B})^{-1/2}, \quad (1)$$

which shrinks the component along each basis row $\mathbf{b}_j$ by $(1 + \kappa \nu_j)^{-1/2}$ and leaves the orthogonal complement of the basis unchanged (Queisler, 2026c). Cohort whitening uses $\mathbf{B} = \mathbf{U}$ and $\boldsymbol{\nu} = \boldsymbol{\lambda}$; pool-basis whitening uses the eigen-decomposition of $\boldsymbol{\Sigma}_{\text{pool}}$, the between-patient covariance of all training patients of the split.

Let $\mathbf{P}_\perp = \mathbf{I} - \mathbf{U}^\top \mathbf{U}$ project off the cohort basis. Queisler (2026e) formed the within-patient complement from the eigen-decomposition $\mathbf{P}_\perp \hat{\boldsymbol{\Sigma}}_W \mathbf{P}_\perp = \mathbf{V}^\top \text{diag}(\boldsymbol{\omega}) \mathbf{V}$ of the cohort’s pooled within-patient covariance, and whitened with $\mathbf{B} = [\mathbf{U}; \mathbf{V}]$ and $\boldsymbol{\nu} = [\boldsymbol{\lambda}; s\tau\boldsymbol{\omega}]$, where τ scales the added trace to $s \text{tr} \hat{\boldsymbol{\Sigma}}_B$ and s is tuned on validation. The rows of $\mathbf{V}$ fill the whole complement of the cohort basis on TCGA-UT, 2,438 directions at five patients, and about 1,085 directions on BRACS, where the number of patch residuals is smaller than the feature dimension.

The *oracle weight* of a unit direction $\mathbf{v}$ is the pool’s between-patient variance along it, $\mathbf{v}^\top \boldsymbol{\Sigma}_{\text{pool}} \mathbf{v}$, as in the oracle spectrum of Queisler (2026d). Three arms keep the cohort block $(\mathbf{U}, \boldsymbol{\lambda})$ unchanged and add a complement block with oracle weights:

$$\text{Wo: } \mathbf{B} = [\mathbf{U}; \mathbf{V}], \quad \nu_{r_B+j} = \mathbf{v}_j^\top \boldsymbol{\Sigma}_{\text{pool}} \mathbf{v}_j, \quad (2)$$

$$\text{Ro: } \mathbf{B} = [\mathbf{U}; \mathbf{R}], \quad \nu_{r_B+j} = \mathbf{r}_j^\top \boldsymbol{\Sigma}_{\text{pool}} \mathbf{r}_j, \quad (3)$$

$$\text{Po: } \mathbf{B} = [\mathbf{U}; \mathbf{Q}_\perp], \quad \mathbf{P}_\perp \boldsymbol{\Sigma}_{\text{pool}} \mathbf{P}_\perp = \mathbf{Q}_\perp^\top \text{diag}(\boldsymbol{\nu}_\perp) \mathbf{Q}_\perp. \quad (4)$$

Wo is the arm the explanation of Queisler (2026e) calls for: its directions are those of the tuned arm Wt, and only the weights change from patch-level to patient-level variance. Ro uses the random orthonormal complement directions $\mathbf{R}$ of the same source study, with the same number of rows as $\mathbf{V}$, and gives them oracle weights; the contrast Wo–Ro measures what the within-patient directions contribute once the weights are correct. Po replaces the added directions by the eigenvectors $\mathbf{Q}_\perp$ of the pool’s between-patient covariance projected off the cohort basis, with their own eigenvalues. It is the best that any complement block can do in the sense that its whitening matrix contains $\mathbf{P}_\perp \boldsymbol{\Sigma}_{\text{pool}} \mathbf{P}_\perp$ exactly, and so it is the ceiling of the construction “cohort block plus complement block”. Its rank equals that of the projected pool covariance, which is the full complement on TCGA-UT and 227 directions on average on BRACS.

The three arms differ in a way that a span measure cannot see. On TCGA-UT, $\mathbf{V}$, $\mathbf{R}$, and $\mathbf{Q}_\perp$ all span essentially the same subspace, the complement of $\mathbf{U}$, and their oracle weights sum to the same trace, the pool variance outside the cohort basis. Wo and Ro nonetheless differ from Po. Their whitening matrices contain only the diagonal of $\mathbf{P}_\perp \boldsymbol{\Sigma}_{\text{pool}} \mathbf{P}_\perp$ in the basis $\mathbf{V}$ or $\mathbf{R}$; the off-diagonal terms in that basis are dropped. By the Schur–Horn theorem, the diagonal of a symmetric matrix in any orthonormal basis is majorized by its eigenvalues (Horn, 1954), so any basis other than the eigenbasis spreads the same variance more evenly over more directions. The

contrast Po–Wo therefore measures the cost of the orientation of $\mathbf{V}$ within the complement, and Wo–Wt the cost of its weights.

The remaining gap RW–Po measures what the block construction itself gives up relative to pool-basis whitening. Po keeps the cohort’s eigenvalues on $\mathbf{U}$, which Queisler (2026d) found to cost only 0.08 points, and drops the covariance between the cohort basis and its complement, $\mathbf{U}\Sigma_{\text{pool}}\mathbf{P}_{\perp}$. The scale κ is tuned as a grid factor divided by the mean eigenvalue (Queisler, 2026c). As in Queisler (2026e), the grid factors are multiplied by the ratio of the arm’s mean eigenvalue to the cohort’s mean eigenvalue, so that every factor applies the same κ to the cohort block as in cohort whitening. No trace parameter s is tuned: the oracle weights fix the size of the added block.

3 Design

The study keeps the setting of Queisler (2026b) on TCGA-UT and of Queisler (2026a) on BRACS. TCGA-UT contributes 30 cancer types and BRACS its seven lesion classes, both with frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), three fixed patient-disjoint splits, and multinomial logistic regression whose regularization strength is selected on validation patient-macro balanced accuracy. For each split and each of ten draws, the cohort of five patients per class with 32 patches each is the one used by all preceding studies of this series, so every classifier trains on 160 patches per class. The study runs at five patients only, where Queisler (2026e) found its only resolved gain; at ten and twenty patients that gain had already vanished.

The cohort basis, its eigenvalues, the within-patient complement, and the random complement directions are recomputed from the same cohort with the same seeds as in the source study, so Wo shares its directions with Wt and Ro with Rt. Table 1 lists the arms. Each fitted arm whitens by Equation 1 and fits an ordinary multinomial logistic classifier on the transformed features; it tunes the scale factor $\kappa \in \{0.1, 1, 10, 100\}$, rescaled as described in Section 2, jointly with the regularization strength on validation patient-macro balanced accuracy, with the grid and tie rules of Queisler (2026c). The reused arms are the exact fits of the source studies on the same cohorts, so every contrast is paired draw by draw.

Per dataset the study fits three arms for each of 30 split-draw pairs, that is 90 whitened fits, each tuning four scale factors and nine regularization values. Selections were written to disk before any test prediction, and test accuracy never entered selection.

4 Evaluation

The endpoint is patient-macro balanced accuracy on the test partition, pooled over splits and draws with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates; all arms share the replicates with the source studies, so every contrast is paired, and intervals are 95 percent percentile intervals.

The primary contrast is Wo5–Wt5, read separately on each dataset. The weight explanation counts as supported on a dataset if oracle weights gain at least one percentage point over the tuned patch-level weights and the paired interval lies entirely above zero. Three bracketing contrasts complete the ladder $\text{RWc} \rightarrow \text{Wt} \rightarrow \text{Wo} \rightarrow \text{Po} \rightarrow \text{RW}$: Wo5–Ro5, Po5–Wo5, and RW5–Po5. The *headroom share* $(x - \text{RWc5})/(\text{RW5} - \text{RWc5})$ states for each arm x how much of the gap between cohort-basis and pool-basis whitening it closes.

Three diagnostics explain the weights. The Spearman correlation between the within-patient eigenvalues ω_j and the oracle weights of the same directions asks whether the patch-level ranking already tracks patient-level strength. The *oracle-implied added trace* $\sum_j \nu_{r_B+j} / \sum_j \lambda_j$ is the value of s at which the added block of Wt would carry the oracle trace, and it can be compared

Arm	Added directions	Added weights	Role
R5	—	—	Ordinary fit on the real cohort (reused)
C5	—	—	Cohort with pool class centres (reused headroom reference)
RWc5	none	—	Cohort whitening, the host (reused)
Wt5	within-patient $\mathbf{V}$	patch-level, tuned s	Expanded cohort whitening (reused)
Rt5	random $\mathbf{R} \perp \mathbf{U}$	patch-level, tuned s	Random-direction control of Wt (reused)
Wo5	within-patient $\mathbf{V}$	oracle, Equation 2	Tests the weight explanation
Ro5	random $\mathbf{R} \perp \mathbf{U}$	oracle, Equation 3	Value of the directions under correct weights
Po5	pool complement $\mathbf{Q}_\perp$	oracle eigenvalues	Ceiling of the block construction
RW5	pool basis	pool eigenvalues	Pool-basis whitening (reused mechanism reference)

Table 1: All fitted arms keep the cohort basis and its eigenvalues and add a complement block weighted by the pool’s between-patient variance; they differ only in the orientation of the added directions. Wo, Ro, Po, and RW read the training pool and are mechanism references, not remedies. The reused arms come from Queisler (2026b), Queisler (2026c), Queisler (2026a), and Queisler (2026e).

with the s selected by Wt. The added pool share $\sum_j \nu_{r_B+j} / \text{tr } \Sigma_{\text{pool}}$ states how much pool variance the added block carries. Four readings were fixed in advance. $\text{Wo} \gg \text{Wt}$ with $\text{Wo} \approx \text{Po}$ would mean that the source study found usable directions and lacked only the weights. $\text{Wo} \approx \text{Wt}$ with $\text{Po} \gg \text{Wo}$ would mean that the within-patient directions are wrong at the resolution whitening needs, even when correctly weighted. $\text{Po} \ll \text{RW}$ would mean that the block construction itself caps the gain. $\text{Ro} \approx \text{Wo}$ would mean that the oracle weights alone carry any gain and the directions do not matter.

5 Results

The results follow the ladder from cohort whitening to pool-basis whitening: first the accuracy of every rung, then the contrasts between adjacent rungs, and finally the diagnostics that explain why the oracle weights change nothing. The reused arms reproduce the values of the source studies exactly, because they are the same fits on the same bootstrap replicates.

5.1 Accuracy along the ladder

The oracle-weighted within-patient arm lands exactly where the tuned patch-level arm does, and the pool’s complement eigenbasis lands close to pool-basis whitening (Table 2). On TCGA-UT, Wo5 reaches 65.18 percent against 65.19 for Wt5, a gain of 0.44 points over cohort whitening (0.27 to 0.61). Po5 reaches 66.86 percent, a gain of 2.12 points (1.78 to 2.45) of the 2.66 points that separate cohort-basis from pool-basis whitening. Random directions with oracle weights gain 0.02 points (−0.00 to 0.05), as the random control with patch-level weights did. BRACS shows the same order with wider intervals: Wo5 gains 0.32 points (−0.31 to 0.94), Ro5 0.05 (−0.02 to 0.21), and Po5 2.33 (0.97 to 3.88) of a 3.01-point gap.

Figure 1 shows the headroom shares side by side. On both datasets the three arms whose added directions come from the cohort or from chance close less than a fifth of the gap, and the arm whose added directions come from the pool closes about four fifths. The step between the two groups is the change of direction, not the change of weight.

Arm	TCGA-UT		BRACS	
	Accuracy	Headroom	Accuracy	Headroom
Real cohort (R5)	62.48 [61.24, 63.65]	—	36.18 [33.46, 38.80]	—
Cohort basis (RWc5)	64.74 [63.52, 65.90]	0	36.49 [33.87, 39.24]	0
Random, patch (Rt5)	64.77 [63.54, 65.93]	—	36.56 [33.91, 39.27]	—
Random, oracle (Ro5)	64.76 [63.53, 65.93]	0.007 [−0.001, 0.018]	36.54 [33.89, 39.26]	0.015 [−0.009, 0.077]
Within, patch (Wt5)	65.19 [63.95, 66.34]	0.169 [0.105, 0.231]	36.94 [34.39, 39.48]	0.149 [−0.140, 0.391]
Within, oracle (Wo5)	65.18 [63.94, 66.32]	0.166 [0.107, 0.228]	36.81 [34.18, 39.46]	0.106 [−0.126, 0.324]
Pool complement (Po5)	66.86 [65.62, 67.99]	0.796 [0.714, 0.880]	38.82 [36.10, 41.27]	0.773 [0.482, 1.031]
Pool basis (RW5)	67.40 [66.14, 68.52]	1	39.50 [36.82, 41.98]	1
Pool centres (C5)	73.14 [71.62, 74.50]	—	44.63 [41.33, 48.05]	—

Table 2: Oracle weights leave the within-patient arm where it was; only the pool’s own complement directions close most of the whitening gap. Accuracy is test patient-macro balanced accuracy in percent at five patients per class; the headroom share is $(x - \text{RWc5})/(\text{RW5} - \text{RWc5})$ and is 0 and 1 for the two whitening references by definition. Intervals are paired 95% bootstrap intervals over test patients and draws. “Cohort basis” and “Pool basis” denote cohort and pool-basis whitening, “Within” and “Random” the within-patient and random complement directions, “patch” and “oracle” the added weights. R5, C5, RWc5, RW5, Wt5, and Rt5 are reused fits (Queisler, 2026b,c,a,e).

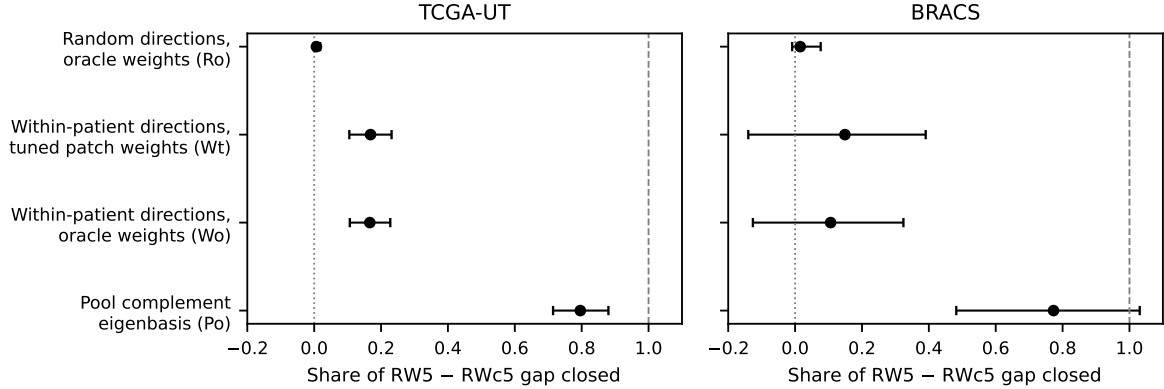


Figure 1: Correcting the orientation of the added directions closes most of the whitening gap; correcting their weights closes no more of it than the tuned patch-level weights. Share of the gap between cohort whitening (dotted line, 0) and pool-basis whitening (dashed line, 1) closed by each arm at five patients per class, with paired 95% bootstrap intervals.

5.2 Ladder contrasts

The primary contrast rejects the weight explanation on both datasets (Table 3). On TCGA-UT, Wo5–Wt5 is -0.01 points, with an interval from -0.11 to 0.10 that excludes not only the one-point threshold but any gain larger than a tenth of a point. The difference is also stable across splits, at -0.09 , 0.09 , and -0.02 points. On BRACS the contrast is -0.13 points (-0.73 to 0.42); its upper bound lies well below the threshold, although the interval is five times wider.

Contrast	What it isolates	TCGA-UT	BRACS
Wo5 – Wt5	Oracle against patch-level weights	-0.01 [$-0.11, 0.10$]	-0.13 [$-0.73, 0.42$]
Wo5 – Ro5	Within-patient against random directions	0.42 [$0.26, 0.59$]	0.27 [$-0.35, 0.88$]
Po5 – Wo5	Pool against within-patient orientation	1.67 [$1.37, 1.99$]	2.01 [$0.74, 3.38$]
RW5 – Po5	Cost of the block construction	0.54 [$0.31, 0.77$]	0.68 [$-0.08, 1.62$]
Wo5 – RWc5	Oracle-weighted within-patient block	0.44 [$0.27, 0.61$]	0.32 [$-0.31, 0.94$]
Ro5 – RWc5	Oracle-weighted random block	0.02 [$-0.00, 0.05$]	0.05 [$-0.02, 0.21$]
Po5 – RWc5	Pool complement block	2.12 [$1.78, 2.45$]	2.33 [$0.97, 3.88$]
RW5 – RWc5	Whole gap	2.66 [$2.33, 2.97$]	3.01 [$1.51, 4.68$]

Table 3: The directions, not the weights, separate the within-patient block from the pool’s complement block. Paired contrasts in percentage points at five patients per class with 95% bootstrap intervals; bold entries are the primary contrast. The first four rows are the steps of the ladder RWc $\rightarrow$ Wt $\rightarrow$ Wo $\rightarrow$ Po $\rightarrow$ RW, apart from the step Wt–RWc reported by Queisler (2026e).

The within-patient directions are better than random ones under the same oracle weights on TCGA-UT, by 0.42 points (0.26 to 0.59), which matches the specificity contrast Wt5–Rt5 of 0.42 points under patch-level weights (Queisler, 2026e). Correct weights therefore neither create nor remove the value of these directions. The largest step is the change of orientation: Po5 exceeds Wo5 by 1.67 points on TCGA-UT (1.37 to 1.99) and by 2.01 points on BRACS (0.74 to 3.38), and it does so in all three splits of both datasets. The last step, RW5–Po5, is 0.54 points on TCGA-UT (0.31 to 0.77), resolved but below the threshold, and 0.68 points on BRACS (-0.08 to 1.62), not resolved.

5.3 Diagnostics

The diagnostics explain why oracle weights add nothing to the within-patient directions (Table 4). The patch-level variances ω_j rank the within-patient directions almost exactly as the pool’s between-patient variance along them does: the Spearman correlation is 0.999 on TCGA-UT and 0.981 on BRACS, averaged over 30 fits. The oracle weights also imply an added trace of 0.296 times the cohort trace on TCGA-UT, which coincides with the best fixed value $s = 0.3$ of the source study, and 0.412 on BRACS, where that study also peaked at $s = 0.3$. Up to the exact shape of the profile, the tuned patch-level arm therefore already carried the oracle weights, both in ranking and in total size.

On TCGA-UT the three added blocks are indistinguishable by every span measure. Each spans the whole complement of the cohort basis and carries 0.30 of the pool’s between-patient variance, which is all the pool variance outside the cohort basis (Queisler, 2026d). Yet Ro5 gains nothing, Wo5 gains 0.44 points, and Po5 gains 2.12 . What distinguishes them is how the same trace is spread over directions, which is the property governed by the Schur–Horn majorization of Section 2: random directions spread it almost evenly, the within-patient directions concentrate it partly, and only the pool’s eigenbasis concentrates it fully. On BRACS the within-patient complement covers $1,085$ of the $2,532$ complement directions but still carries 0.401 of the 0.423

	TCGA-UT			BRACS		
	Wo5	Ro5	Po5	Wo5	Ro5	Po5
Added directions	2,438	2,438	2,440	1,085	1,085	227
Added pool share	0.302	0.301	0.302	0.401	0.181	0.423
Oracle-implied added trace	0.296	0.296	0.296	0.412	0.186	0.435
Spearman(ω , oracle)	0.999	—	—	0.981	—	—
Selected factor 0.1	0	0	0	12	18	4
Selected factor 1	5	23	0	7	6	2
Selected factor 10	21	7	1	10	4	10
Selected factor 100	4	0	29	1	2	14

Table 4: On TCGA-UT all three added blocks carry the same pool variance in the same subspace; they differ only in how it is distributed over directions. Added directions and shares are means over 30 fits at five patients per class. The added pool share is the fraction of the pool’s between-patient trace carried by the added block, and the oracle-implied added trace is that block’s trace divided by the cohort trace. The selection rows count the scale factor, before rescaling, chosen in the 30 fits.

pool share outside the cohort basis, whereas random directions of the same number carry 0.181. The pool’s complement covariance there has rank 227 only, so Po5 concentrates the same variance on a fifth as many directions as Wo5.

The selected scale follows the concentration. On TCGA-UT, Ro5 keeps factor 1 in 23 of 30 fits, the choice of cohort whitening, and Wo5 selects factor 10 in 21 of 30 fits, as Wt5 did (Queisler, 2026e). Po5 selects the largest factor, 100, in 29 of 30 fits. Because the factors are anchored to the cohort block, this choice applies to both blocks a κ one hundred times that of the factor cohort whitening usually selects, and validation still prefers it over every smaller factor. On BRACS, Po5 selects factor 100 in 14 and factor 10 in 10 of 30 fits, whereas Wo5 and Ro5 mostly select the smaller factors 0.1 and 1.

6 Discussion

Of the four readings fixed in Section 4, the results select the second on both datasets: $Wo \approx Wt$ and $Po \gg Wo$. The explanation offered by Queisler (2026e), that the within-patient directions fail because they carry patch-level instead of patient-level weights, is rejected. Oracle weights change the accuracy by at most a tenth of a point on TCGA-UT, and the diagnostics show why: the patch-level variances already rank the within-patient directions as the pool does, and validation already chose the oracle trace. The first reading is excluded because the weights were not missing. The third is excluded because Po closes about four fifths of the gap. The fourth is excluded on TCGA-UT, where the within-patient directions beat random ones by the same 0.42 points under either weighting, and is unresolved on BRACS.

The bottleneck is the orientation of the added directions within the right subspace. On TCGA-UT, the within-patient complement, the random complement, and the pool’s complement eigenbasis span the same space and carry the same pool variance, yet their gains are 0.44, 0.02, and 2.12 points. Whitening by Equation 1 contracts each basis direction by its own weight and cannot represent variance that is shared between directions. The pool’s between-patient variance outside the cohort basis is concentrated on comparatively few directions, and only a basis aligned with those directions contracts them strongly while leaving the rest of the complement, and the class signal in it, largely intact. In any other basis, the Schur–Horn theorem guarantees that the same variance appears more evenly spread, so whitening must either

contract many directions moderately or few directions too weakly. The selected scale factors fit this picture: the concentrated Po block tolerates the strongest contraction on the grid, the partly concentrated Wo block a moderate one, and the nearly flat Ro block none beyond cohort whitening. The within-patient directions are informative, since they beat random directions, but their orientation reflects how patches vary within a patient, and that variation does not share the principal axes of patient-to-patient variation.

The result closes the line of cohort-only span remedies opened by Queisler (2026d) and Queisler (2026e). The headroom outside the cohort basis is real and large, 2.12 of 2.66 points on TCGA-UT with the pool’s complement eigenbasis, but it requires the principal axes of between-patient variation outside the cohort’s span. Those axes are a property of the patient population. A cohort of five patients per class determines at most $C(G - 1)$ of them, its patch residuals determine a different set of axes, and correct weights on the wrong axes recover nothing more. A remedy along these lines would have to estimate the population’s between-patient axes from patients outside the labelled cohort, for example from unlabelled slides of further patients, since patient means require no labels.

The block construction itself costs little. Po keeps the cohort’s eigenvalues on the cohort basis and drops the covariance between that basis and its complement, and still reaches 80 percent of pool-basis whitening on TCGA-UT. The remaining 0.54 points are resolved but below the practical threshold. Correcting the cohort eigenvalues alone is worth 0.08 points (Queisler, 2026d), so the remainder plausibly lies in the dropped cross-covariance or in the coarse scale grid; neither was isolated here.

7 Limitations

All fitted arms read the between-patient covariance of the full training pool, so none of them is a remedy, and the study answers only where the bottleneck lies. It runs at five patients per class only; the source study found no gain at ten or twenty patients to explain, but the ladder at those counts was not measured.

The oracle weights are the diagonal of the pool covariance in the within-patient basis. This is the natural reading of “patient-level weights” for the whitening of Equation 1, which can only attach one weight to each basis direction, but it is not the only possible correction. A rotation within the within-patient complement that uses cohort-only information was not tried, and the conclusion that the orientation is the bottleneck applies to the within-patient eigenbasis as defined in Section 2. The Schur–Horn argument explains why a non-eigenbasis must spread the variance, but the eigenvalue profiles of the three blocks were not recorded, so the degree of concentration is inferred from the accuracy and the selected scale, not measured.

The scale grid limits Po. It selects the largest factor in 29 of 30 TCGA-UT fits, so a wider grid could raise Po5 and shrink the step RW5–Po5 further; the reported Po5 is a lower bound on the ceiling of the block construction, which strengthens rather than weakens the contrast Po5–Wo5. On BRACS, Po adds only 227 directions against 1,085 for Wo and Ro, so its contrast with them mixes orientation with a difference in the number of contracted directions, and all BRACS intervals are wide. The arms reuse the three evaluation splits of every earlier study in this series. The results are conditional on frozen Virchow2 features, multinomial logistic regression, 160 patches per class, and the two cohorts studied.

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